EMPLOYEE DATA ANALYSIS

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TABLE OF CONTENTS

3 OVERVIEW

4 INTRODUCTION

5 DATA EXPLORATION

6 DATA PREPROCESSING

7 DATA VISUALIZATION

12 CONCLUSION

OVERVIEW

THE GOAL OF THIS ANALYSIS IS TO UNDERSTAND THE STRUCTURE OF THE DATA, EXPLORE THE RELATIONSHIPS BETWEEN DIFFERENT VARIABLES, AND VISUALIZE KEY PATTERNS WITHIN THE DATASET.

WE AIM TO IDENTIFY IMPORTANT TRENDS SUCH AS EMPLOYEE DISTRIBUTION ACROSS DEPARTMENTS, SALARY PATTERNS, AND OTHER RELEVANT CHARACTERISTICS THAT HELP BETTER UNDERSTAND THE DATASET.

INTRODUCTION

DATA ANALYSIS PLAYS AN ESSENTIAL ROLE IN EXTRACTING MEANINGFUL INSIGHTS FROM RAW DATASETS. IN THIS PROJECT, WE ANALYZE AN EMPLOYEE DATASET USING PYTHON LIBRARIES SUCH AS PANDAS, MATPLOTLIB, AND SEABORN.

THE ANALYSIS PROCESS INCLUDES SEVERAL STAGES:

1. EXPLORING THE DATASET
2. CLEANING AND PREPROCESSING THE DATA
3. VISUALIZING THE DATA TO IDENTIFY PATTERNS AND TRENDS

WE AIM TO TRANSFORM RAW DATA INTO USEFUL INFORMATION THAT CAN SUPPORT DECISION-MAKING.

DATA EXPLORATION

WE EXPLORE THE DATASET TO UNDERSTAND ITS STRUCTURE AND CONTENTS

THE EXPLORATION PROCESS INCLUDES:

- VIEWING THE FIRST ROWS OF THE DATASET USING HEAD()
- CHECKING THE DATASET SHAPE (NUMBER OF ROWS AND COLUMNS)
- IDENTIFYING COLUMN NAMES AND DATA TYPES
- CHECKING FOR MISSING VALUES
- UNDERSTANDING STATISTICAL SUMMARIES OF NUMERICAL COLUMNS

DATA PREPROCESSING

DATA PREPROCESSING IS A CRUCIAL STEP BEFORE PERFORMING ANALYSIS. RAW DATASETS OFTEN CONTAIN MISSING VALUES, DUPLICATE ROWS, OR UNNECESSARY COLUMNS THAT NEED TO BE CLEANED.

THE PREPROCESSING STEPS PERFORMED INCLUDE:

- HANDLING MISSING VALUES
- REMOVING UNNECESSARY COLUMNS
- CONVERTING DATA TYPES WHEN NEEDED
- REMOVING DUPLICATES
- PREPARING THE DATASET FOR ANALYSIS

THESE STEPS ENSURE THAT THE DATASET BECOMES CLEAN AND SUITABLE FOR ANALYSIS

DATA VISUALIZATION

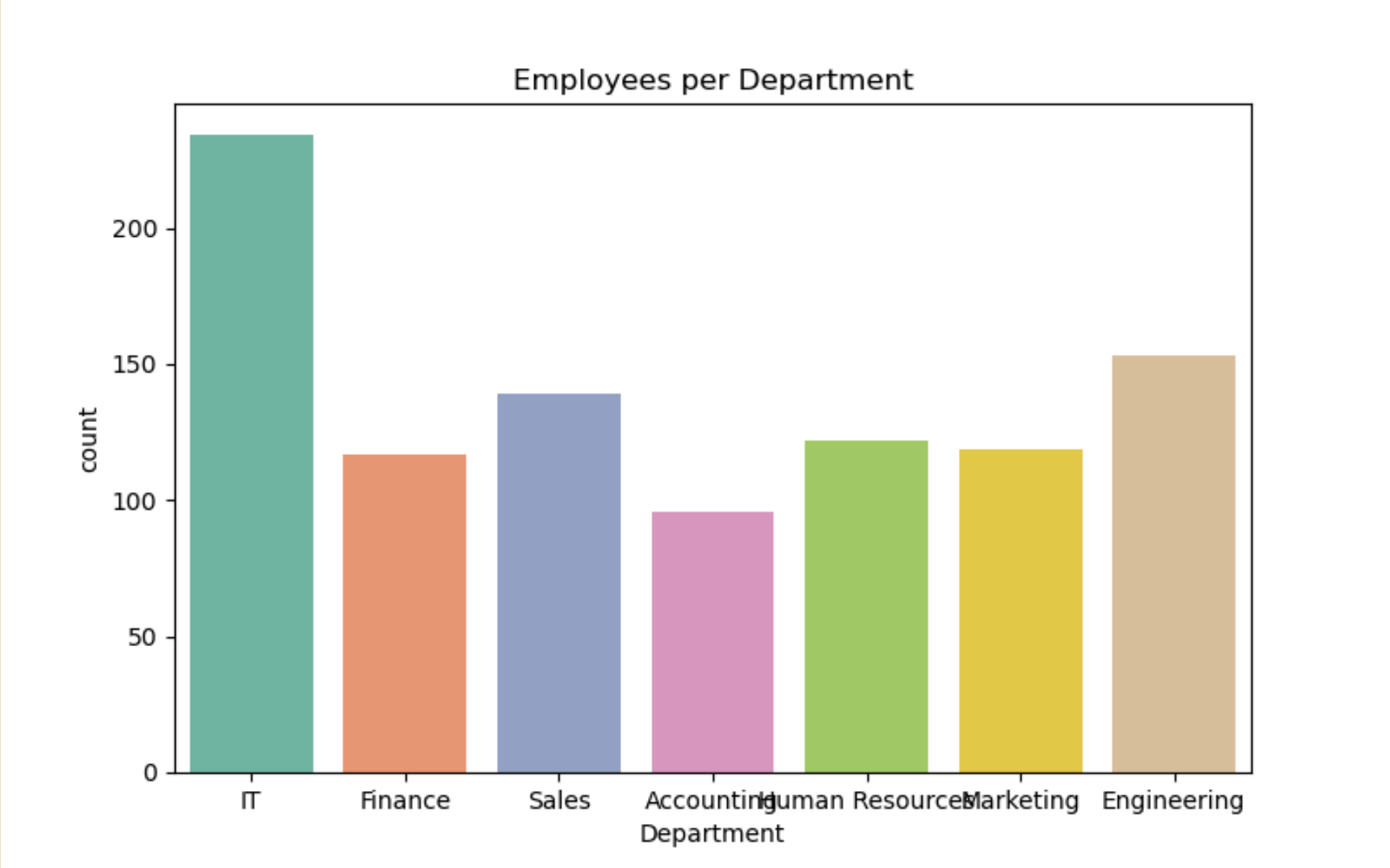
DATA VISUALIZATION HELPS REPRESENT DATA IN A GRAPHICAL FORMAT, MAKING IT EASIER TO IDENTIFY PATTERNS, RELATIONSHIPS, AND TRENDS

IN THIS PROJECT, WE USED MATPLOTLIB AND SEABORN TO CREATE DIFFERENT TYPES OF VISUALIZATIONS, SUCH AS:

- BAR CHARTS
- HISTOGRAMS
- PIE CHARTS
- COUNT PLOTS

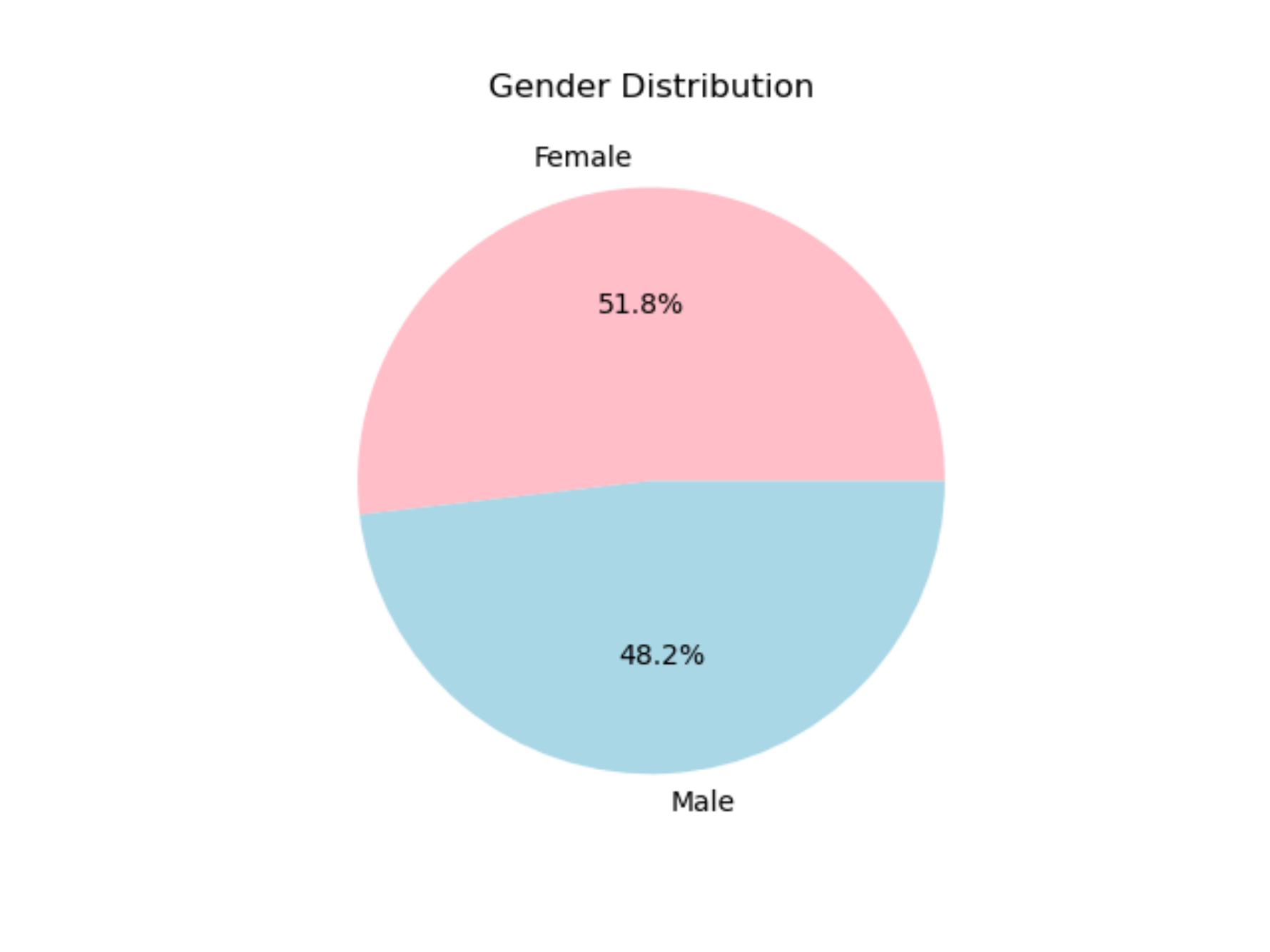
BAR CHARTS

THIS BAR CHART SHOWS THE NUMBER OF EMPLOYEES IN EACH DEPARTMENT. THE IT DEPARTMENT HAS THE HIGHEST NUMBER OF EMPLOYEES



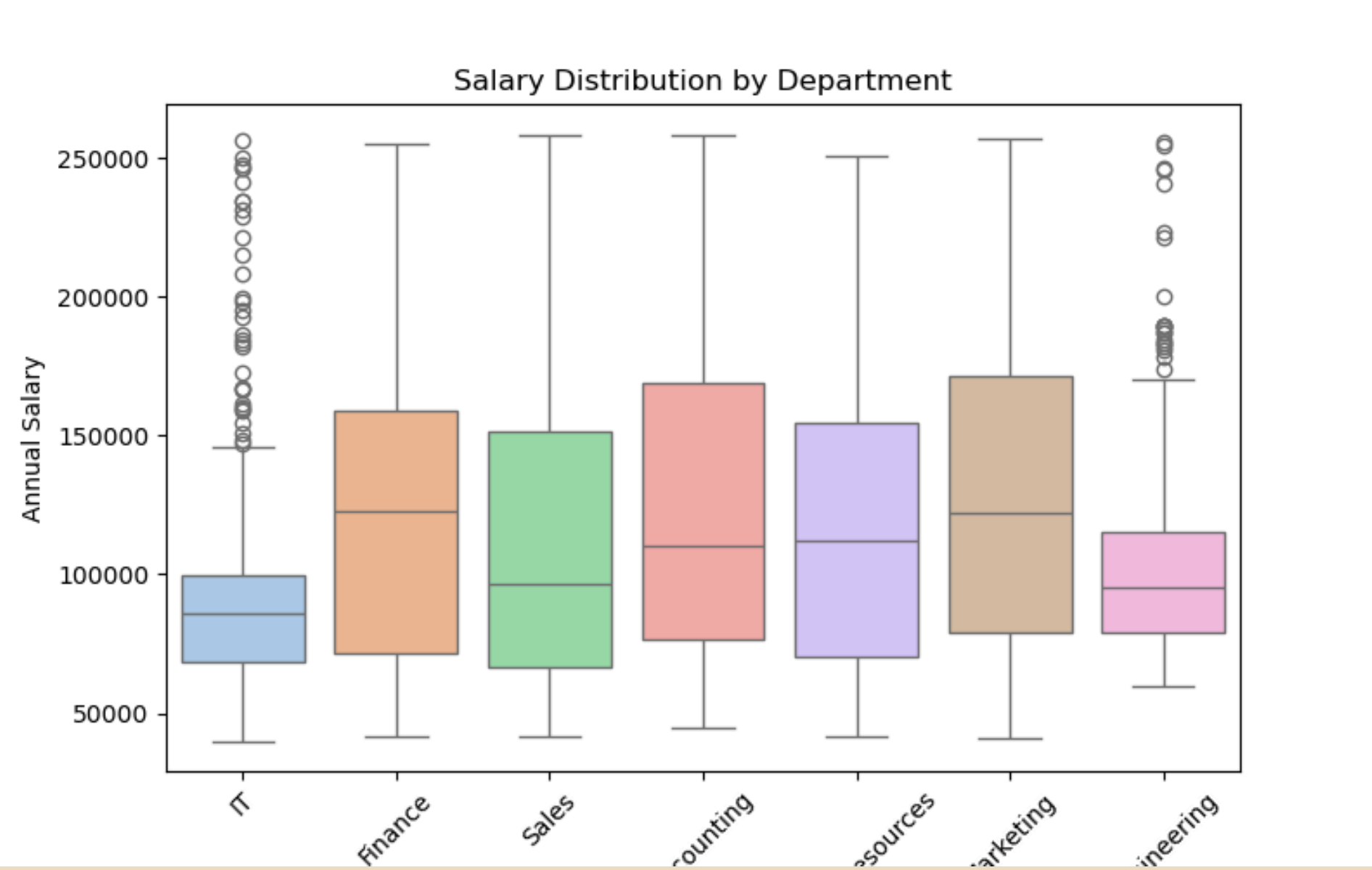
PIE CHART

THIS PIE CHART SHOWS THE DISTRIBUTION OF EMPLOYEES BY GENDER. THE PERCENTAGE OF FEMALE EMPLOYEES IS SLIGHTLY HIGHER (51.8%) COMPARED TO MALE EMPLOYEES (48.2%). THIS INDICATES THAT THE GENDER REPRESENTATION IN THE DATASET IS RELATIVELY BALANCED, WITH A SMALL MAJORITY OF FEMALE EMPLOYEES.



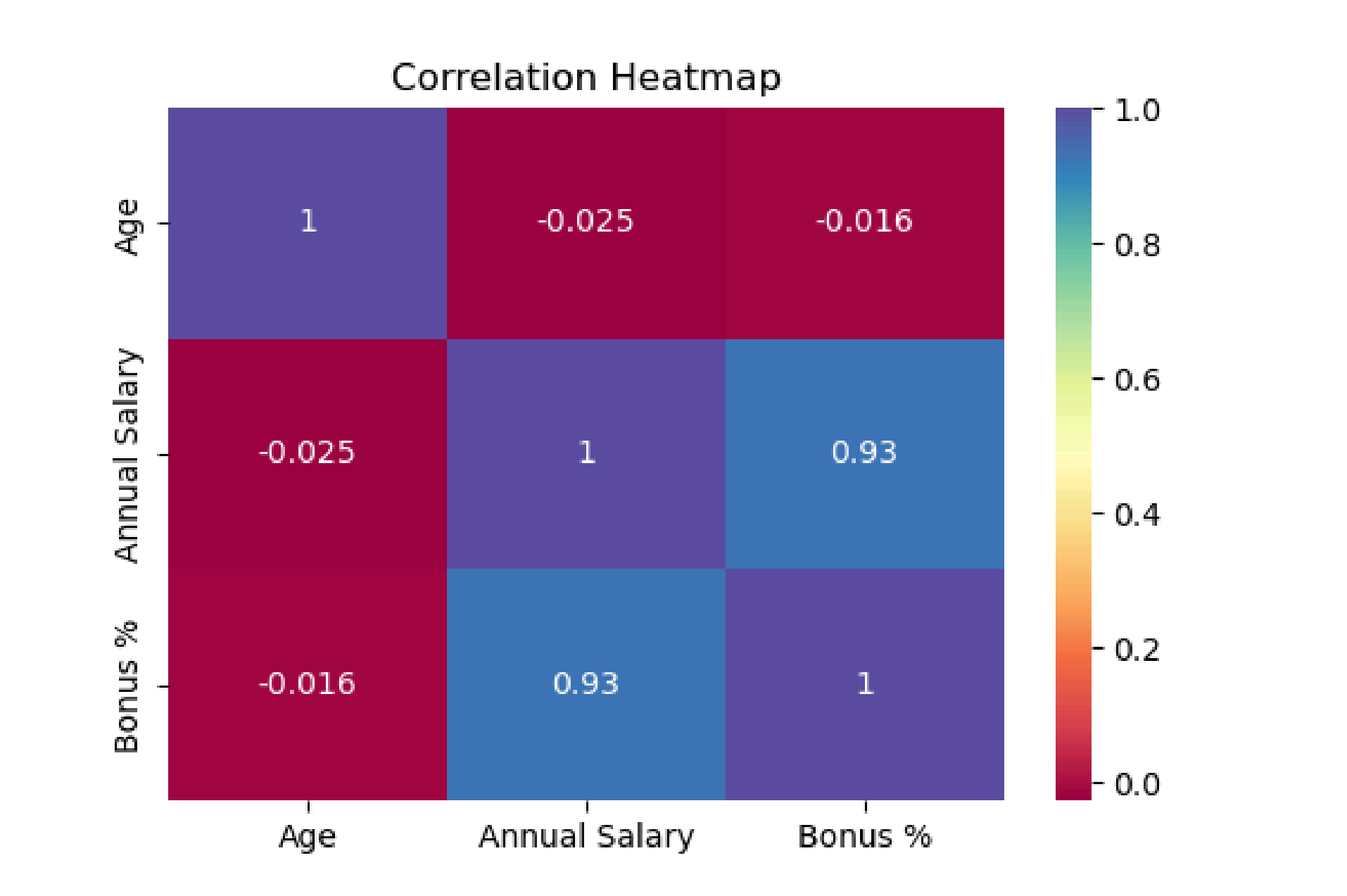
BOX PLOT

THIS BOX PLOT SHOWS HOW SALARIES ARE DISTRIBUTED ACROSS DIFFERENT DEPARTMENTS. IT HIGHLIGHTS THE VARIATION IN SALARIES, INCLUDING THE MEDIAN, RANGE, AND POTENTIAL OUTLIERS IN EACH DEPARTMENT. SOME DEPARTMENTS SHOW A WIDER SALARY RANGE, INDICATING GREATER VARIATION IN EMPLOYEE SALARIES. OVERALL, THE VISUALIZATION HELPS COMPARE SALARY PATTERNS BETWEEN DEPARTMENTS AND IDENTIFY DEPARTMENTS WITH HIGHER OR MORE VARIABLE SALARIES.



HEATMAP

THIS HEATMAP SHOWS THE RELATIONSHIP BETWEEN AGE, ANNUAL SALARY, AND BONUS %. THERE IS A STRONG POSITIVE CORRELATION (0.93) BETWEEN ANNUAL SALARY AND BONUS %, MEANING HIGHER SALARIES ARE ASSOCIATED WITH HIGHER BONUSES. AGE SHOWS VERY WEAK CORRELATION WITH THE OTHER VARIABLES.



CONCLUSION

IN CONCLUSION, THIS ANALYSIS PROVIDED USEFUL INSIGHTS INTO THE EMPLOYEE DATASET. THROUGH DATA EXPLORATION, PREPROCESSING, AND VISUALIZATION, WE WERE ABLE TO UNDERSTAND THE DATASET STRUCTURE AND IDENTIFY IMPORTANT PATTERNS.

THE VISUALIZATIONS HELPED HIGHLIGHT KEY TRENDS WITHIN THE DATA, MAKING IT EASIER TO INTERPRET THE RESULTS. DATA PREPROCESSING ALSO ENSURED THE DATASET WAS CLEAN AND RELIABLE FOR ANALYSIS.

OVERALL, THE PROJECT DEMONSTRATES HOW DATA ANALYSIS TECHNIQUES CAN TRANSFORM RAW DATA INTO MEANINGFUL INSIGHTS THAT SUPPORT BETTER UNDERSTANDING AND DECISION-MAKING.